Code No: 123BP **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, October - 2020
DATA STRUCTURES
(Common to CSE, IT)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Distinguish between circular linked list and doubly linked list by giving one example for each.
b) Write a C function for deleting an integer element from the singly linked list of integer elements. Assume that the singly linked list of integers is already created. [7+8]

2.a) Write a C function for inserting an integer element into a circular queue of integers. Assume array representation for the circular queue.
b) Convert the following infix expression into postfix form a+b*c-e/f . [8+7]

3. Write a non recursive procedure for the inorder and post order traversal of a binary tree. Assume that the binary tree of elements is already created. [15]

4.a) Write heap sort algorithm for sorting a list of integers in ascending order.
b) What is the time complexity of heap sort in the worst case? [8+7]

5.a) Compare the performance of AVL tree with binary search tree.
b) Write the Knuth-Morris-Pratt pattern matching algorithm. [8+7]

6.a) Explain with an example any one method used for representing sparse matrix.
b) Write a C function for concatenating two singly linked linear lists of characters. Assume that the two singly linked lists of characters are already created. [7+8]

7. Write a C program to implement stack using singly linked list. [15]

8.a) Explain with an example the adjacency matrix representation of a graph also mention its complexities.
b) Write a recursive procedure for the DFS of a graph. [8+7]

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